Yixuan Yang

yixuan.yang@duke.edu | www.yixuanyang.com

Duke UNIVERSITY

Education

Duke University *08/2024 – Present* *Durham, United States*
Ph.D. in Electrical and Computer Engineering, Advisor: ***Prof. Rishi Kamaleswaran*** GPA: 3.96/4.0

Southern University of Science and Technology (SUSTech) *08/2020 – 06/2024* *Shenzhen, China*
B.E. in Computer Science, Academic Advisor: ***Prof. Xin Yao (Fellow, IEEE)*** GPA: 3.53/4.0

- Distinguished Graduate, Department of Computer Science (Top 1%, 2 out of 245 students)
- Guo Xie Bi Rong Fellowship (Top 4%); SUSTech Outstanding Undergraduate Alumnus, Class of 2024

University of California, Berkeley (UCB) *08/2022 – 01/2023* *Berkeley, United States*
Exchange Undergraduate Student GPA: 4.0/4.0

Research Interest

Instruction Tuning & Reasoning-Enhanced LLM, Representation Learning, Foundation Models for EHR, ML for Combinatorial Optimization, Evolutionary Algorithms, Humanoid Robot Manipulation, Multi-Agent Systems.

Research Experiences

Instruction Tuning Foundational Model for Trajectory Modeling & Reasoning *08/2025 – Present*
Duke University – Kamaleswaran Lab ***Advised by Prof. Rishi Kamaleswaran***

- Developed reasoning-enhanced, instruction-tuned Large Language Models to model acute illness trajectories and automate clinical planning from Electronic Health Record (EHR) data. Built a framework that transforms longitudinal patient data into real-time actionable insights for clinical decision support.

Automated Lab: Scientific Lab Manipulation driven by Humanoid Robot *08/2024 – 03/2025*
Duke University – General Robotics Lab ***Advised by Prof. Boyuan Chen***

- Engineered a high-fidelity sim-to-real learning platform and physical manipulation toolkit for the Unitree G1 humanoid robot to automate complex tasks in scientific laboratory environments.
- Developed a robust manipulation framework in Nvidia Isaac Gym utilizing a hybrid approach of Deep Reinforcement Learning (PPO) and traditional heuristics to enable precise, contact-rich control.

GasHunter: Empowering Efficient Gas Source Localization by Collaborative Robots *05/2023 – 06/2024*
Tsinghua University - UC Berkeley Institute (Visiting Research) ***Advised by Prof. Xinlei Chen***

- Proposed *GasHunter*, a collaborative multi-robot system for gas source localization in patchy plume environments. Combined Langevin-inspired planning with dynamic multi-agent scheduling to balance exploration and exploitation.
- Achieved 20%+ higher success rate and 30%+ path efficiency over SOTA baselines in both simulation and real-world testbed.

Machine Learning-Based Problem Reduction for Large-Scale UFLP Optimization *01/2023 – 06/2024*
Southern University of Science and Technology ***Advised by Prof. Xin Yao (Fellow, IEEE)***

- Proposed a ML-based framework for large-scale UFLP optimization by transferring knowledge from smaller instances.
- Proposed the *Local Adjacent Vector (LAV)* strategy, reducing instance size by up to 86.8% with <1% solution quality loss. Achieved significant acceleration and improved convergence across four benchmark meta-heuristics.

Publications

- Yuhan Cheng*, Xuecheng Chen*, **Yixuan Yang**, Haoyang Wang, Jingao Xu, Chaopeng Hong, Susu Xu, Xiaoping Zhang, Yunhao Liu, and Xinlei Chen. **SniffySquad: Patchiness-Aware Gas Source Localization with Multi-Robot Collaboration**. ***ACM Transactions on Sensor Networks, Jan 2026***. DOI: 10.1145/3786599.
- Yuhan Cheng*, Xuecheng Chen*, **Yixuan Yang**, Haoyang Wang, Yuxuan Liu, and Xinlei Chen. **Poster: Olfactory Sensing in Turbulent Airflow via Collaborative Robots**. In ***Proceedings of the 25th International Workshop on Mobile Computing Systems and Applications (ACM HotMobile)***, 2024.
- Shuaixiang Zhang, **Yixuan Yang**, Hao Tong, and Xin Yao. **Learning-Based Problem Reduction for Large-Scale Uncapacitated Facility Location Problems**. In ***IEEE Congress on Evolutionary Computation (CEC)***, 2024.

Academic Service & Technical Skills

Government Service: Student Researcher in **North Carolina AI Leadership Council (Tech, Data, & Infra team) -** hosted by the **NC Office of AI and Policy** to support government agencies in AI-related decision-making.
Teaching Assistant: DECISION 520Q/618: Data Science for Business, Duke Fuqua School of Business (08/2024 – 10/2024).
Reviewer: IEEE Journal of Selected Topics in Signal Processing, Special Issue on Intelligent Robotics (2023).
Programming: Python, JavaScript, React, C/C++, Java, SQL, LaTeX
Frameworks & Tools: PyTorch, Hugging Face Ecosystem, CUDA, MIMIC-IV / EHR Databases, Nvidia Isaac Gym/Genesis, Git, Linux, Raspberry Pi, Arduino, Fusion 360, UWB Systems